

When Deep Learning Does Not Help Multi-view Diffusion: Truncated SVD and Nyström Beat the Neural Machinery

Cautionary study — Centre Borelli

Abstract

A parametric network (Deep Diffusion Maps, DDM) could in principle improve Multi-view Diffusion Trajectories (MDT)—whose embedding is the SVD of a product of per-view diffusion operators—along four axes: *accuracy*, *cost*, *out-of-sample* extension, and *end-to-end* trainability. We report a negative result. Accuracy is capped by a ceiling argument: a Gram-matching network’s global optimum is the truncated SVD itself (Eckart–Young), so it cannot exceed the SVD of the same operator—confirmed empirically. On cost, a truncated SVD on the operator’s sparse action matches the network’s accuracy at $\sim 300\times$ less compute. On out-of-sample—DDM’s actual motivation—a single run appears to favour Nyström, but over five seeds Nyström and the encoder *tie within a large run-to-run variance*: out-of-sample is a wash. The lone axis no argument or baseline forecloses, end-to-end geometry learning, fails in practice—an optimisation/conditioning failure (the contrastive loss stalls and the operator spectrum flattens), not the collapse it superficially resembles. Several early apparent “wins” were artifacts of weak baselines—a poor single random trajectory, and dense eigendecomposition where a truncated iterative SVD belongs—which we document explicitly, because they manufacture progress that does not exist.

1 What we tested

An MDT operator is a length- t product $\mathbf{W} = \mathbf{W}_t \cdots \mathbf{W}_1$, each step a convex mixture of per-view row-stochastic transitions $\mathbf{W}_s = \sum_v a_{sv} \mathbf{P}_v$; its embedding is the truncated SVD of $\mathbf{W}$. MDT is transductive and, with a dense product, costs $O(N^3)$. DDM offers a parametric encoder. The combination is appealing: a network might give cheaper, out-of-sample, or more accurate embeddings. We tested whether it does, and—crucially—*against the baselines a referee would demand*, not the convenient ones.

Four axes, and what closes each. A network’s embedding can differ from the classical SVD on four axes, and each is closed by an argument or by the correct baseline. *Accuracy* is closed by a ceiling: a network trained to match the embedding’s Gram matrix, $\|\mathbf{F}\mathbf{F}^\top - \mathbf{G}\|^2$, has its global optimum at the truncated SVD of $\mathbf{G}$ (Eckart–Young¹),

¹ $\mathbf{G}$ is symmetric PSD with eigendecomposition $\mathbf{G} = \sum_l \sigma_l^2 \mathbf{u}_l \mathbf{u}_l^\top$ ($\sigma_1^2 \geq \sigma_2^2 \geq \cdots$). As $\mathbf{F}$ ranges over $N \times k$ matrices, $\mathbf{F}\mathbf{F}^\top$ ranges over all rank- k PSD matrices, and Eckart–Young gives $\min_{\text{rank } k} \|\mathbf{M} - \mathbf{G}\|_F$

so it cannot exceed the SVD of the same operator. *Cost*: the correct baseline is **truncated svds** on the operator’s sparse action (a **LinearOperator** applying $\mathbf{W}$ by sparse matrix–vector products, never forming $N \times N$). *Out-of-sample*: the correct baseline is **Nyström** extension. Note **svds** produces only a table of values over the training rows—it cannot map a new point—so it is not a competitor here; Nyström is the SVD’s own out-of-sample extension. *End-to-end*: making the embedding a trainable layer under a downstream objective is the one axis no baseline forecloses—but it fails in practice (§3.4). A **features-only** k -means control isolates how much the operator contributes at all.

2 Setup: what each scenario changes

Baselines. DDM (single view) learns an embedding by Gram-matching to a *fixed* spectral target,

$$\min_{\Gamma} \|\Pi^{1/2} \Gamma^\top \Gamma \Pi^{1/2} - (\mathbf{A}^{2t} - \sqrt{\pi} \sqrt{\pi}^\top)\|_F^2, \quad (1)$$

with $\mathbf{A} = \mathbf{D}^{-1/2} \mathbf{K}_\alpha \mathbf{D}^{-1/2}$ symmetric and π its stationary distribution. MDT (multi-view) instead builds the *asymmetric* operator $\mathbf{W} = \mathbf{W}_t \cdots \mathbf{W}_1$, $\mathbf{W}_s = \sum_v a_{sv} \mathbf{P}_v$, $\mathbf{P}_v = \mathbf{D}_v^{-1} \mathbf{K}_v$, and reads its truncated SVD $\mathbf{W} = \mathbf{U} \Sigma \mathbf{V}^\top$, $\Psi = \mathbf{U}_{:,1:k} \Sigma_{1:k}$ (transductive).

Unified target. Let π be the stationary distribution of $\mathbf{W}$ ($\mathbf{W}^\top \pi = \pi$, $\Pi = \text{diag}(\pi)$) and $\mathbf{A} = \Pi^{1/2} \mathbf{W} \Pi^{-1/2}$ its symmetrisation, whose top singular vector is $\sqrt{\pi}$. The diffusion-map embedding is the top- k factor of

$$\mathbf{G} = \mathbf{A} \mathbf{A}^\top - \sqrt{\pi} \sqrt{\pi}^\top = \sum_{l \geq 1} \sigma_l^2 \mathbf{u}_l \mathbf{u}_l^\top. \quad (2)$$

This $\mathbf{G}$ is the MDT analogue of DDM’s $\mathbf{A}^{2t} - \sqrt{\pi} \sqrt{\pi}^\top$: since $\mathbf{W}$ is asymmetric, the symmetric power $\mathbf{A}^{2t}$ is replaced by $\mathbf{A} \mathbf{A}^\top$. Each scenario changes only what is *fixed* vs *learned* and the objective (Table 1).

at $\mathbf{M}^* = \sum_{l \leq k} \sigma_l^2 \mathbf{u}_l \mathbf{u}_l^\top$, i.e. the top- k truncated eigendecomposition. Hence the optimal $\mathbf{F}^* = \mathbf{U}_{:,1:k} \text{diag}(\sigma_{1:k})$ up to a right orthogonal rotation—exactly the SVD embedding. A parametric $\mathbf{F} = f_\theta(x)$ can approach this optimum but, being a subset of all $N \times k$ matrices, can never beat it.

Table 1: The four scenarios as modifications of DDM/MDT. $S = \text{diag}(\sqrt{\pi})$; $\mathbf{w}_*$ is the operator row of a new point x_* against the training set. Scenarios 1–3 keep $\mathbf{W}$ fixed and only change how its SVD embedding is read out or extended; scenario 4 makes $\mathbf{W}$ itself trainable, removing the fixed target and collapsing.

Scenario	fixed / learned	objective	change vs DDM	change vs MDT
1. Accuracy (neural SVD)	$\mathbf{W}$ fixed; learn f_θ	$\min_\theta \ S f_\theta f_\theta^\top S - \mathbf{G}\ _F^2$	$\mathbf{A}^{2t} \rightarrow \mathbf{A}\mathbf{A}^\top$; multi-view input	SVD readout $\rightarrow$ NN learns it
2. Cost (truncated SVD)	$\mathbf{W}$ fixed; no network	top- k svds of $x \mapsto \mathbf{W}x$	drop the network	dense $O(N^3) \rightarrow$ matrix-free
3. Out-of-sample	$\mathbf{W}$ fixed; learn $x_* \mapsto \psi$	$f_\theta(x_*)$ vs Nyström $\mathbf{w}_*^\top \mathbf{V}$	use f_θ for new points	add extension (none in MDT)
4. End-to-end	learn $\phi_v, a: \mathbf{W}(\theta)$	$\min_\theta \mathcal{L}_{\text{contr}}(\mathbf{W}(\theta))$	$\mathbf{G} \rightarrow$ trainable $\mathbf{G}(\theta)$	operator fixed $\rightarrow$ learned

3 Results

3.1 Accuracy: the encoder only matches the SVD (a ceiling)

The most seductive early finding was a parametric “neural SVD” beating classical eig by up to +0.20 AMI. It was an artifact of the *decomposition*, not the operator or the network. MDT embeds by plain truncated SVD of its operator; the $\sqrt{\pi}$ symmetrisation is DDM’s out-of-sample machinery, not MDT’s, and plays no role here. The +0.20 baseline instead used a $\sqrt{\pi}$ -weighted *dense* eig. Holding the decomposition fixed at truncated svds (Table 2), every operator variant—a random trajectory, an unsupervised best-of-eight, the oracle best-of-eight, and the neural reproduction—scores within 0.03 (0.77–0.81). Trajectory *selection* adds nothing once fair: a spectral-energy selector ties a random trajectory (regret to the oracle 0.01–0.03), because the near-uniform spectrum makes the eight trajectories nearly equivalent. The network merely reproduces the SVD it approximates (neural $\approx$ eig_{best}). The operator itself still helps—features-only k -means is far weaker (0.45 on MNIST-10k)—so diffusion contributes, but neither the selection nor the network does.

3.2 Cost: a sparse truncated SVD is $\sim 300\times$ faster

The “linear-time beats $O(N^3)$ ” claim compared the network against *dense* eigendecomposition—a baseline nobody uses for a top- k embedding. The correct method, truncated svds on the same sparse operator action, is sub-second at $N = 10,000$ and *equally accurate*, while the network needs ~ 90 s (Table 3). The neural method is not a fast approximation; it is a slow one.

3.3 Out-of-sample (DDM’s motivation): a wash within run-to-run noise

This is the use case that justified involving DDM: MDT is transductive, so embedding new points needs *some* extension, and svds provides none. The contest is the learned encoder against the training-free Nyström extension of

the same operator. A single-run comparison of a *Gram-matching* encoder (Table 4) appears to favour Nyström, but a *multi-seed* study (Table 5) shows this is not real: across five seeds, Nyström, a regression encoder on raw features, and a regression encoder on the affinity row all *tie within a large run-to-run variance* ($\sigma \approx 0.06$ –0.10 on the small/heterogeneous sets). Any single-seed “win” or “loss”—including an earlier +0.12 for the encoder on MSRC—is inside that noise.

Two controls explain the non-result. (i) A sample-size sweep gave a *flat* gap over $N_{\text{train}} \in [200, 1400]$, ruling out sample size. (ii) The affinity-row encoder reproduces Nyström almost exactly—as expected, since with that input the optimal map *is* Nyström’s linear projection. Where heterogeneous-view data does hurt (Wikipedia), a per-view ablation shows the culprit is *fusion*, not the extension method: the text view alone clusters at 0.51, the image view at 0.05, and fusing them *dilutes* to 0.37 for every method. The honest verdict: **Nyström is never beaten**. The encoder—whether Gram-matching or regression, both multi-seeded—*ties* Nyström where features are informative (MSRC, Handwritten: within ± 0.01 –0.07) and *loses* where they are not (Wikipedia: Gram encoder 0.29 vs 0.38, gap beyond noise). It never wins; out-of-sample is a wash in Nyström’s favour, and all methods need the train-set SVD.

3.4 End-to-end: learning the geometry fails (and not the way one expects)

Letting the network learn the per-view kernels and fusion weights under a contrastive objective collapses clustering quality, well below a fixed random operator (Table 6). The intuitive story—“the geometry collapses to a constant, gaming the loss”—is *wrong*, as instrumenting the training shows. Over 200 steps on Caltech101-7: (a) the contrastive loss barely moves (58.36 $\rightarrow$ 58.09, $< 1\%$), so the objective is *not* being minimised—nothing is “gamed”; (b) the operator’s effective rank *rises* from ~ 1 to ~ 850 (of $N=1474$) and the learned features *spread* (std 1.9 $\rightarrow$ 7.8)—the opposite of a rank-one / constant collapse. The real failure is one of *optimisation and conditioning*: gradient descent

Table 2: The “+0.20 win” was a decomposition artifact. MDT embeds by plain truncated SVD of its operator $\mathbf{W}$ (the $\sqrt{\pi}$ symmetrisation is DDM’s out-of-sample machinery, not MDT’s, and is not used here). With the decomposition held fixed at truncated **svds**, every operator variant—a random trajectory, an unsupervised (spectral-energy) best-of-eight, the oracle (AMI-selected) best-of-eight, and the neural reproduction on the fair operator—scores within 0.03 (0.77–0.81). Trajectory *selection* buys nothing once fair (energy $\approx$ random; regret to the oracle 0.01–0.03), and the network merely reproduces the SVD (neural $\approx$ eig_{best} on the same, energy-selected operator). The operator still beats features-only k -means (0.45 on MNIST-10k), so diffusion contributes—but neither selection nor the network does. The original +0.20 came from a $\sqrt{\pi}$ -weighted *dense* eig, now removed.

Dataset	classes	features only	eig (random) [†]	eig_{best} (fair) [‡]	eig_{best} (oracle)	neural (fair) [§]
MNIST-4	4	0.707	0.789	0.783	0.808	0.782
MNIST-10k	10	0.450	0.786	0.787	0.798	0.790
ALOI	100	0.737	0.773	0.767	0.787	0.795

All embeddings are plain truncated **svds** of the MDT operator (no $\sqrt{\pi}$). [†] mean over 8 random trajectories. [‡] best of 8 by top- k spectral energy (unsupervised, deployable). oracle = best of 8 by AMI (uses labels). [§] neural on the same energy-selected (fair) operator as eig_{best} (fair); it reproduces **svds** (within ± 0.03 run-to-run).

Table 3: Scaling on MNIST-10k. Truncated **svds** on the sparse operator action dominates both the strawman (dense **eigh**) and the neural network on time, at equal accuracy.

N	wall-clock time			accuracy AMI (svds $\approx$ neural)
	dense eigh	svds	neural	
2 000	4.3 s	0.07 s	18.5 s	0.74
4 000	14.6 s	0.11 s	40.0 s	0.79
8 000	98.0 s	0.30 s	71.3 s	0.81
10 000	—	0.31 s	89.4 s	0.81

Table 4: Out-of-sample AMI, *single run*: a Gram-matching encoder vs Nyström. The apparent gaps are *within* the run-to-run variance quantified in Table 5; see text.

Dataset	Nyström	deep (Gram-match)
MSRC-v5	0.689	0.642
Handwritten	0.885	0.862
Wikipedia	0.505	0.348
UCI	0.836	0.839
OutdoorScene	0.482	0.482
Caltech101-7	0.522	0.555

through the kernel $\rightarrow$ product $\rightarrow$ contrastive chain stalls, and the small drift it does make *flattens the operator’s singular spectrum*, erasing the spectral gap that a top- k embedding relies on, so AMI falls. The stall is robust to tuning—sweeping the learning rate over $[0.001, 0.2]$ and to 600 steps leaves the loss flat ($< 1\%$) in *every* case—which is the signature of a flat landscape, not a step-size problem. A landscape analysis pins the root cause: the t -step diffused operator has entries $\approx 1/N$ ($|\overline{W_{ij}}| = 6.8\text{e-}4$), so the contrastive logits $\exp(W)$ are *uniform*—softmax(W) has mean max-probability 0.00068, exactly $1/N$. The objective therefore sits in a dead-flat region, with vanishing gradients (fusion weights $\|\nabla_a\| \approx 10^{-5}$); no learning rate helps because there is essentially no gradient. Confirming this, introducing a logit temperature ($\exp(\tau W)$,

Table 5: Out-of-sample AMI over 5 *seeds* (mean $\pm$ std). Nyström, a regression encoder on raw features, and one on the affinity row all *tie within the std*: out-of-sample is a wash. The affinity encoder matches Nyström by construction. Single-seed effects (cf. Table 4) are noise.

Dataset	Nyström	deep (feat.)	deep (aff.)
MSRC-v5	0.648 $\pm$ 0.091	0.652 $\pm$ 0.061	0.652 $\pm$ 0.045
Wikipedia	0.303 $\pm$ 0.101	0.255 $\pm$ 0.073	0.305 $\pm$ 0.106
Handwritten	0.864 $\pm$ 0.032	0.862 $\pm$ 0.032	0.858 $\pm$ 0.032

Table 6: End-to-end learning of the geometry (learned kernels + weights) fails, worse than fixed operators. AMI. The failure is optimisation/conditioning (flat loss, spectrum flattening), not the rank-one collapse the numbers might suggest—see text.

Dataset	fixed+random	fixed+learned wts	learned geom.
MSRC-v5	0.682	0.470	0.328
Caltech101-7	0.535	0.410	0.049
OutdoorScene	0.570	0.463	0.005
UCI	0.822	0.914	0.751

$\tau=20$) unsticks the loss (-8.7% vs -0.5%) and lifts AMI (0.07 $\rightarrow$ 0.13)—so the failure is a *logit-scale* problem, fixable by sharpening but not by tuning lr. Even unstuck it stays below the fixed-operator baseline, so end-to-end remains not worth it; we now simply know *why*. (Separately, in the *imitative* setting the $\sqrt{\pi}$ weighting and a batch-normalised head are needed to avoid the $\mathbf{F}=0$ saddle of the Gram loss.)

Mitigations, and why they do not rescue the method. The wash-out has an obvious countermeasure—diffuse less—so we test two. (a) *Short product*: shrink the trajectory length t , keeping W farther from the (uniform) stationary operator. (b) *Laziness gate*: replace each step by a lazy walk $S'_s = g_s S_s + (1 - g_s)\mathbf{I}$ with a learnable $g_s = \sigma(\theta_s) \in [0, 1]$, penalised by $\lambda \sum_s g_s$ —a differentiable surrogate for choosing t , since the integer t carries no

Table 7: t -sweep of three methods on the four collapse datasets (AMI). *fixed* MDT = best-of-8 length- t operator; *imit.* = the $\sqrt{\pi}$ -Gram encoder trained to reproduce that fixed operator; *end-to-end* = learns the operator under the contrastive objective (Scenario 4, 5-seed mean). Fixed and imit are *stable* across t —imit reproduces fixed at every length—while end-to-end *collapses* as t grows (well-connected kNN graphs over-mix past one hop), except UCI (one dominant view). Learning the *encoder* of a fixed operator works; learning the *operator* under the contrastive loss does not. (Weak-random and gated- $\lambda=1$ variants in text.)

Dataset	method	$t=1$	$t=2$	$t=3$	$t=4$
MSRC-v5	fixed MDT	.704	.736	.728	.720
	imit. (Gram)	.712	.724	.701	.706
	end-to-end	.533	.309	.277	.239
Caltech101-7	fixed MDT	.547	.588	.619	.620
	imit. (Gram)	.643	.545	.647	.644
	end-to-end	.234	.094	.070	.085
OutdoorScene	fixed MDT	.629	.633	.606	.598
	imit. (Gram)	.579	.561	.522	.525
	end-to-end	.340	.001	.098	.177
UCI	fixed MDT	.846	.834	.830	.840
	imit. (Gram)	.841	.810	.840	.828
	end-to-end	.867	.837	.763	.730

gradient. The t -sweep is striking: $t=1$ is *best on every dataset* and AMI falls monotonically with t (Table 7), often by a wide margin (MSRC 0.53 at $t=1$ vs 0.24 at $t=4$)—the learned model always wants *minimal* diffusion, so MDT’s multi-step machinery never helps it. This is the mixing-vs-hops tension: the well-connected kNN graphs mix so fast that even two hops over-blur classes that are essentially a one-hop property. Yet three caveats hold. (i) *Passive*: the contrastive loss still barely moves (-0.1 to -1.1%), so any gain is the operator resting near a structured single-hop/identity regime, not the optimiser learning. (ii) *At least one hop is needed*: gating toward the identity ($\lambda=1$, $\bar{g} \rightarrow 0.16$, *less* than one hop) collapses OutdoorScene (0.072) where a single full hop ($t=1$) scores 0.34. (iii) *It never beats a fair baseline*: against best-of-eight trajectories (5 seeds), even the best learned setting ($t=1$) *loses* on 3/4—MSRC (0.53 vs 0.71), Caltech (0.23 vs 0.62), OutdoorScene (0.34 vs 0.60)—and merely *ties* on the well-conditioned UCI (0.867 vs 0.854, bands overlap). The apparent UCI “win” over *fixed+random* was only against the single-trajectory strawman. The mitigations thus confirm the wash-out mechanism and reduce learned geometry to a single-hop operator, but do not make it beat a fairly-tuned fixed operator; the negative result stands.

Is the collapse a build artifact? A controlled kernel ablation. Our end-to-end model deviates from both references in a way worth isolating: it makes the kernel learnable through a per-view feature projection (Linear($d, 32$), randomly initialised), whereas MDT builds

Table 8: Controlled dense-vs-sparse kernel ablation ($t=4$, $\tau=20$; identity init, max-min bandwidth, all else fixed; AMI). “init” = untrained operator. Both kernels’ untrained operators match fixed MDT—*neither collapses*; training degrades both far below init; dense $\approx$ sparse. The failure is the objective, not the kernel.

Dataset	fixed MDT	sparse kNN		dense	
		init	trained	init	trained
MSRC-v5	0.720	0.624	0.457	0.658	0.492
OutdoorScene	0.598	0.562	0.335	0.530	0.312
UCI	0.840	0.769	0.506	0.802	0.390

kernels on *raw* features and DDM’s network outputs the embedding directly—neither projects kernel inputs. Two consequences follow. (a) The *random projection*, not the operator, drove the original collapse: re-initialising the projection to the identity (raw-feature distances) and adding a logit temperature τ removes it—the *untrained* operator then matches fixed MDT on every dataset (Table 8, “init”). (b) Dense vs. sparse kernel is a *wash*: holding init, bandwidth, t , τ and seed fixed and toggling only the operator’s kNN mask, the two move AMI only modestly and with no consistent sign (across 12 configurations), and *neither collapses at init nor escapes collapse when trained*. Note the two kernels are *not* numerically interchangeable in general: on the low-dimensional views the tight max-min bandwidth makes the dense Gaussian decay to ≈ 0 off-neighbourhood (the kNN mask keeps 100% of the mass), but on the high-dimensional views ($d=512, 576$) the mask drops real off-neighbourhood mass (only $\sim 5\text{--}9\%$ kept). That the outcome is a wash *even though the kernels genuinely differ* is the stronger point. The robust, kernel-independent conclusion: with a correctly-built operator whose untrained state already equals fixed MDT, gradient-training on the contrastive objective still *degrades* clustering (23/24 controlled runs)—the objective, not the kernel or the conditioning, is misaligned with clustering. The faithful way to couple MDT and DDM is therefore the imitative encoder (Gram-matching a fixed MDT embedding), not end-to-end kernel learning.

3.5 What does survive (and does not need a network)

Two minor positive findings remain, both classical and both cheap with *svds*; both held up under multi-seed scrutiny better than the OOS or collapse claims did. (i) *Consensus (robust)*: averaging the diffusion targets over trajectories beats any single trajectory on all datasets, and over 5 seeds the gain *exceeds* the run-to-run std, from $+0.01$ (MSRC, $0.70 \rightarrow 0.72$) to $+0.14$ (100Leaves, $0.77 \rightarrow 0.90$); it also removes the selection step. (ii) *Selection criterion (weak)*: the silhouette index correlates with downstream AMI somewhat better than the Calinski–Harabasz index

of the original MDT, but the effect is modest—over seeds, $\rho = 0.55 \pm 0.20$ vs 0.40 ± 0.24 , an advantage that sits *within* the large run-to-run variance, not the clean $2 \times$ a single run suggested. Neither finding needs the parametric machinery; both are operations on the SVD embedding.

A better internal-quality index. Pursuing the selection question further, we compared three label-free trajectory-selection indices on the canonical kernel over a pool of 30 random trajectories $\times 3$ seeds (Table 9): MDT’s contrastive Internal Quality index (Eq. 19–20 of the MDT paper), silhouette, and a *spectral energy* index—the fraction of the operator’s singular-value energy carried by its top- k modes. The spectral index is the only reliable ranker (mean $\rho = +0.23$ vs -0.01 and -0.05 ; the other two are uncorrelated with quality on this protocol, so silhouette’s edge over Calinski–Harabasz is itself protocol-dependent) and gives the lowest selection regret on 5/7 structured datasets. It is *not* universal: on heterogeneous-view Wikipedia-test (one near-useless view) it selects a high-energy near-degenerate trajectory (regret 0.32) where the contrastive index succeeds—the same fusion pathology that dilutes MDT there. The gain is *MDT-side, not DDM*: DDM’s defining $\sqrt{\pi}$ weighting is inert here, because the diffused operator’s near-uniform rows make $\pi \approx$ uniform and $\mathbf{A} \approx \mathbf{W}$ (spectral energy of $\mathbf{A}$ equals that of $\mathbf{W}$ on every dataset tested). One property—near-uniform diffused operators—thus explains three failures at once: the end-to-end stall (§3.4), the weakness of the contrastive index, and the inertness of the $\sqrt{\pi}$ weighting.

Table 9 uses the SVD embedding dimension $d=k$. Raising it is a *mirage*: at $d=64$ the selector metrics *improve* (mean energy ρ : $+0.23 \rightarrow +0.53$, regret $0.12 \rightarrow 0.08$) yet the *deployed* AMI (oracle–regret) *falls* ($0.54 \rightarrow 0.52$ and lower at $d=128$), because embedding into more coordinates of a flat-spectrum operator lowers the oracle itself—a fourth face of the same near-uniformity, and a reminder that a shrinking regret can track a collapsing ceiling rather than a better result. $d=k$ is optimal.

(iii) *Operator hyperparameters do not give a deployable gain.* Tuning the operator itself (kernel bandwidth—global vs. self-tuning local scaling—graph degree k_{nn} , and normalisation) has *oracle* headroom (best grid cell $+0.06$ AMI over the paper kernel on MSRC), but it is unreachable without labels: silhouette, which ranks trajectories acceptably, mis-ranks these configurations, preferring near-degenerate high-silhouette partitions (on UCI a 0.80-silhouette cell scores only 0.17 AMI), so unsupervised selection *regresses below* the untuned baseline (-0.01 on MSRC). We report this because an early version, benchmarked against a kernel that dropped the self-loop (lazy-walk) term, showed large apparent gains ($+0.18$)—another strawman baseline, now corrected. Consensus remains the only deployable operator-level gain.

Table 9: Unsupervised MDT trajectory selection on the canonical kernel: Spearman ρ with oracle AMI and selection regret (oracle–selected AMI; lower is better), over 30 random trajectories $\times 3$ seeds, on structured datasets (oracle AMI ≥ 0.35). Indices: MDT’s contrastive Internal Quality Q_c , silhouette (sil), and the **spectral energy** index (**en**). Spectral energy is the only positive ranker and lowest-regret on 5/7; it fails on heterogeneous-view Wikipedia-test.

Dataset	Spearman ρ			selection regret		
	Q_c	sil	en	Q_c	sil	en
MSRC-v5	−.25	−.17	.49	.072	.097	.043
Yale	−.12	−.59	.11	.073	.131	.064
BBCSport	.13	−.04	.60	.125	.164	.019
Caltech101-7	−.12	.11	.36	.047	.467	.043
Handwritten	.24	.05	.19	.050	.365	.209
UCI	−.28	−.30	.36	.369	.404	.121
Wikipedia-test	.33	.61	−.47	.000	.000	.315
mean	−.01	−.05	.23	.105	.233	.116

Table 10: Clustering AMI under the MDT paper’s exact protocol, which our pipeline reproduces (AD/ID/MVD on 100Leaves match Table 4 within ~ 0.01). MDT consensus leads on all three, by a modest $+0.04$ – $+0.08$.

Dataset	AD	ID	MVD	MDT-Rand	MDT-cons.
MSRC-v5	0.643	0.616	0.630	0.625	0.686
100Leaves	0.801	0.733	0.814	0.656	0.894
Caltech101-7	0.570	0.440	0.565	0.590	0.609

4 Reproduction, external baselines, and scale

The gaps a first reading flags are closed. *Reproduction*: replicating the MDT paper’s exact protocol—kNN kernel with $k_{nn} = \lceil \log N \rceil$ and max-of-row-min bandwidth, knee-selected diffusion times, **svds** embedding, AMI over 20 KMeans runs, full data—our pipeline reproduces their Table 4: on 100Leaves we obtain AD 0.801, ID 0.733, MVD 0.814 against the paper’s 0.800, 0.738, 0.824. *Field baselines (fair protocol)*: under that same protocol the MDT consensus embedding leads AD/ID/MVD/MDT-Rand on all three shared datasets (Table 10)—but only by $+0.04$ to $+0.08$. (A weaker re-implementation we used initially inflated this gap by crippling the baselines; the fair numbers below are modest.) *Scale*: on synthetic data the sparse-operator truncated **svds** runs at $N=10^5$ in ~ 27 s where a dense decomposition is infeasible (10^{10} entries, $O(N^3)$). *Multi-seed*: every result above carries error bars. (Scope: the scale test is synthetic—compute-feasibility, not accuracy at scale.)

5 Lessons

- **The embedding’s ceiling is the SVD.** A parametric network can at best match the top- k subspace of its target operator; it cannot exceed it. Any reported excess is a baseline artifact.
- **Strawman baselines manufacture progress.** A bad single trajectory and dense `eigh` made the network look both more accurate and faster than it is. Truncated `svds` on the sparse operator action, and a features-only control, dissolve the apparent gains.
- **Out-of-sample is a wash, and single runs lie.** Over five seeds, Nyström and both encoder variants tie within a large variance (Table 5); the single-seed “wins” and “losses” (including a +0.12 for the encoder on MSRC) are noise. No extension is fundamentally ahead, and all need the train-set SVD.
- **Verify the mechanism, not just the number.** The end-to-end “collapse” is not a collapse at all—instrumenting it shows a stalled loss and a *flattening* spectrum, not the rank-one degeneracy the AMI suggested. Asserted mechanisms must be measured.

6 Conclusion

For embedding multi-view diffusion operators, the right tool is a sparse truncated SVD: it is faster than dense decomposition, faster than the neural network by two orders of magnitude, and equally accurate. The parametric and end-to-end variants of DDM×MDT add cost (and, when learning the geometry, instability) without benefit. We report this so the next attempt starts from the correct baselines.

References

- [1] S. García-Heredia, Á. Fernández, C. M. Alaíz. *Deep Diffusion Maps*. arXiv:2505.06087, 2025.
- [2] G. Debaussart-Joniec, A. Kalogeratos. *Multi-view diffusion geometry using intertwined diffusion trajectories*. arXiv:2512.01484, 2025.